Caleb Woldemedhine

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EDUCATION

University of Denver Denver, CO
B.S. in Computer Science (AI & ML Concentration) Sept 2023 - Mar 2027
Minors: Mathematics and Statistics
GPA: 3.66
Relevant Coursework: Database Organization & Management, Topics in Artificial Intelligence (Machine Learning & LLMs), Applied Math for Data Science and AI, Machine Learning for IoT and Embedded Devices, Linear Algebra, Introduction to Probability

EXPERIENCE

Palo Alto Networks Santa Clara, CA
Software Engineering Intern June 2026 - Present
- Architected an LLM visualization service in a pub/sub backend that profiles data, selects chart components, and formats agent outputs into frontend JSON contracts with a deterministic fallback chain for 100% UI renderability.
- Refactored visualization system into a single-spec catalog architecture, built a benchmarking harness achieving 92% chart-selection accuracy across 500+ scenarios, and migrated the LLM engine from Anthropic to Vertex AI.
- Engineered a 7-worker concurrent E2E test harness automating 73 multi-step Pub/Sub and HTTP flows in QA, featuring mid-run JSON checkpointing, inline JWT refresh, queue purging, and direct result sync to Google Sheets.

Frederick S. Pardee Institute for International Futures Denver, CO
Data Technician Sept 2025 - Present
- Cleaned and preprocessed international education datasets using Python and pandas, standardizing country-year coverage, resolving missing or inconsistent values, and running quality checks to prepare tables for database imports and website updates.
- Built a Python import script to convert cleaned Excel datasets into SQL-ready inserts, automate formatting and loading workflows, and verify imported values against the Data Dictionary for accurate database integration.

Tempus Automation Highlands Ranch, CO
Software Engineering Intern Jun 2025 - Aug 2025
- Trained model using ML.NET and evaluated a C# model to automate deck design generation from computer vision inputs, using a custom dataset of materials, dimensions, pricing rules, and design constraints.
- Built a relational SQL database to store inventory, pricing logic, and model training data, then integrated the trained model into a web platform for real-time design recommendations and dynamic cost estimates.

Denver Center of African Art Denver, CO
Web Development Intern Jun 2024 - Sept 2024
- Built a responsive website using HTML, CSS, and JavaScript and integrated Stripe donation payments with receipts and donor record tracking.

Michael Auto Sales Denver, CO
IT & Data Assistant Sept 2022 - Jan 2024
- Organized profit, expense, and inventory data using Excel, Python, SQL, and DealerCloud to track business performance and maintain accurate vehicle records.

PROJECTS

Stock Prediction and Analysis Tool
- Built a stock analysis and forecasting website with a Python backend and JavaScript frontend that gives users an interactive dashboard with price forecasts, sentiment insights, trading signals, and model accuracy results powered by machine learning.

DuDraw AI Coder
- Built a full-stack AI website where users enter a prompt and a large language model generates DuDraw code that turns their ideas into interactive drawings and simple games.

NYC COVID-19 Spread Simulation
- Created an interactive COVID-19 spread simulation with adjustable controls such as lockdowns, patient zero, speed and real-time epidemic curve visualization.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Next.js, R, Haskell
Libraries & Technologies: PyTorch, TensorFlow, Keras, Scikit-learn, Prophet, Pandas, Plotly, NumPy, Excel, Tableau, REST APIs, AWS, ML.NET, Docker, Git, Linux, JSON

LEADERSHIP

National Society of Black Engineers Denver, CO
President Jun 2025 - Present
- Raised $40,000 to send 35 students to the NSBE National Convention and organized a multi-company hiring fair, leading employer outreach, sponsorships, and event operations to expand student career opportunities.